

COMSATS UNIVERSITY ISLAMABAD

ATTOCK CAMPUS



Assignment 04

Name	Hasnain Ahmad
Registration No	FA23-BSE-035
Submitted to	Sir Kamran
Date	22-05-2026
Subject	MAD

Question 1:

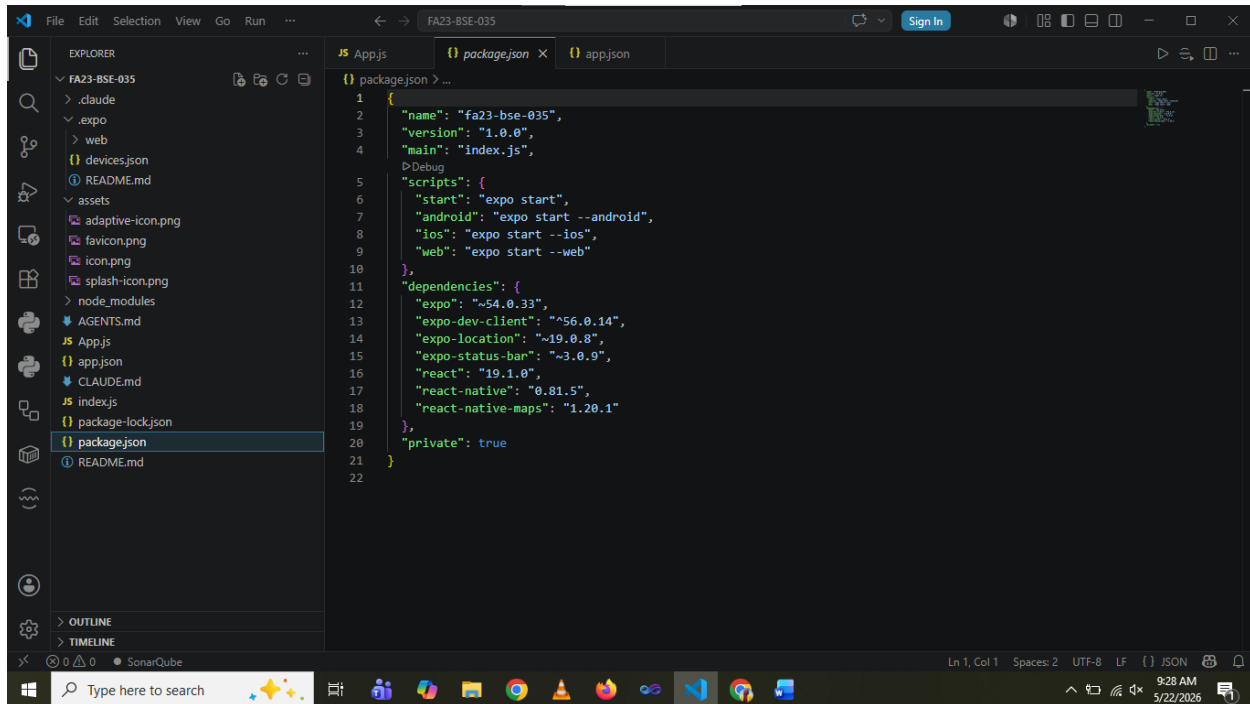
Index.js:

```
1 import { registerRootComponent } from 'expo';
2
3 import App from './App';
4
5 // registerRootComponent calls AppRegistry.registerComponent('main', () => App);
6 // It also ensures that whether you load the app in Expo Go or in a native build,
7 // the environment is set up appropriately
8 registerRootComponent(App);
9
```

Package-lock.json:

```
1 {
2   "name": "fa23-bse-035",
3   "version": "1.0.0",
4   "lockfileVersion": 3,
5   "requires": true,
6   "packages": {
7     "": {
8       "name": "fa23-bse-035",
9       "version": "1.0.0",
10      "dependencies": {
11        "expo": "~54.0.33",
12        "expo-dev-client": "^56.0.14",
13        "expo-location": "~19.0.8",
14        "expo-status-bar": "~3.0.9",
15        "react": "19.1.0",
16        "react-native": "0.81.5",
17        "react-native-maps": "1.20.1"
18      },
19      "node_modules/@0no-co/graphql.web": {
20        "version": "1.2.0",
21        "resolved": "https://registry.npmjs.org/@0no-co/graphql.web/-/graphql.web-1.2.0.tgz",
22        "integrity": "sha512-1IHy9Tr63gE1YCR5Idjx8UREz1s0kFhyd3b8LCxyqjhkic6iGAzTOx3zP1fCwFR87tsh/4Pa9cNt",
23        "license": "MIT",
24        "peerDependencies": {
25          "graphql": "^14.0.0 || ^15.0.0 || ^16.0.0"
26        },
27        "peerDependenciesMeta": {
28          "graphql": {
29            "optional": true
30          }
31        }
32      }
33    }
34  }
35}
```

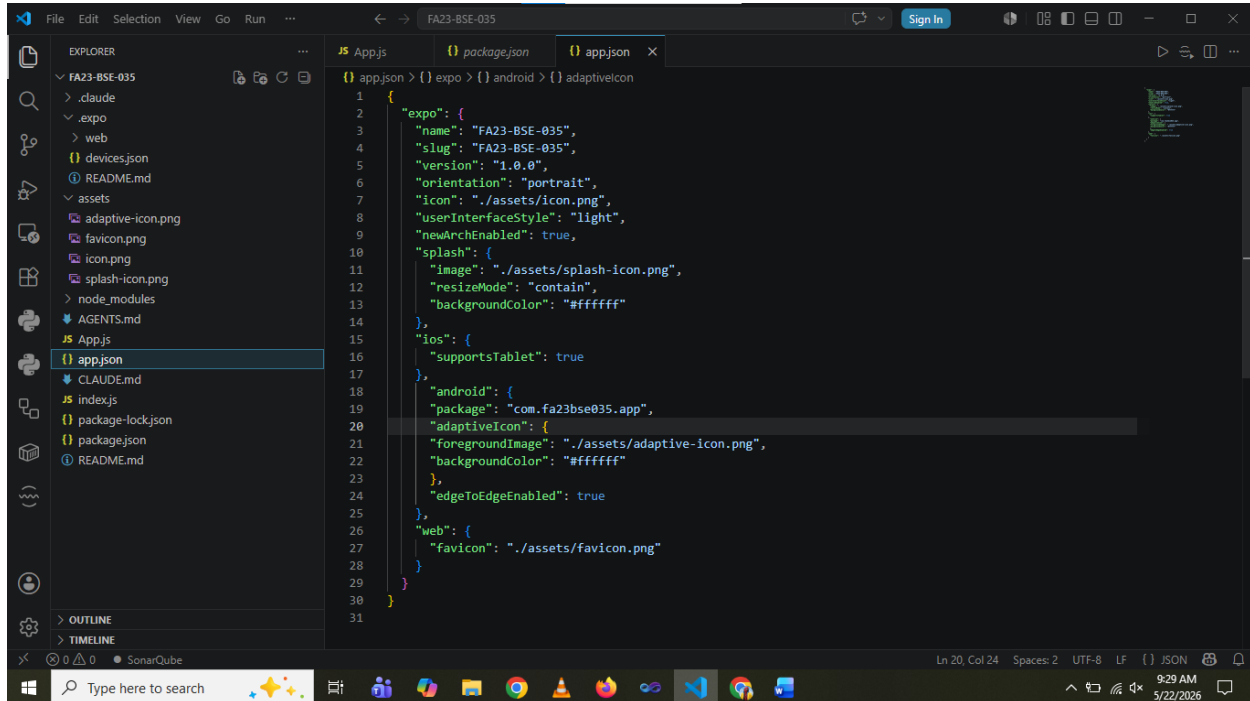
Package. Json:



The screenshot shows the Visual Studio Code editor with the 'package.json' file open. The Explorer sidebar on the left shows the project structure for 'FA23-BSE-035', including files like '.claude', '.expo', 'web', 'devices.json', 'README.md', 'assets', 'adaptive-icon.png', 'favicon.png', 'icon.png', 'splash-icon.png', 'node_modules', 'AGENTS.md', 'App.js', 'app.json', 'CLAUDE.md', 'index.js', 'package-lock.json', and 'package.json'. The 'package.json' file is selected and its content is displayed in the main editor. The code defines the application's metadata, scripts, dependencies, and private status.

```
1 {
2   "name": "fa23-bse-035",
3   "version": "1.0.0",
4   "main": "index.js",
5   "scripts": {
6     "start": "expo start",
7     "android": "expo start --android",
8     "ios": "expo start --ios",
9     "web": "expo start --web"
10  },
11  "dependencies": {
12    "expo": "~54.0.33",
13    "expo-dev-client": "~56.0.14",
14    "expo-location": "~19.0.8",
15    "expo-status-bar": "~3.0.9",
16    "react": "19.1.0",
17    "react-native": "0.81.5",
18    "react-native-maps": "1.20.1"
19  },
20  "private": true
21 }
```

App. json:



The screenshot shows the Visual Studio Code editor with the 'app.json' file open. The Explorer sidebar on the left shows the project structure for 'FA23-BSE-035', including files like '.claude', '.expo', 'web', 'devices.json', 'README.md', 'assets', 'adaptive-icon.png', 'favicon.png', 'icon.png', 'splash-icon.png', 'node_modules', 'AGENTS.md', 'App.js', 'app.json', 'CLAUDE.md', 'index.js', 'package-lock.json', and 'package.json'. The 'app.json' file is selected and its content is displayed in the main editor. The code defines the application's configuration, including the expo configuration, splash screen, iOS and Android settings, and the adaptive icon.

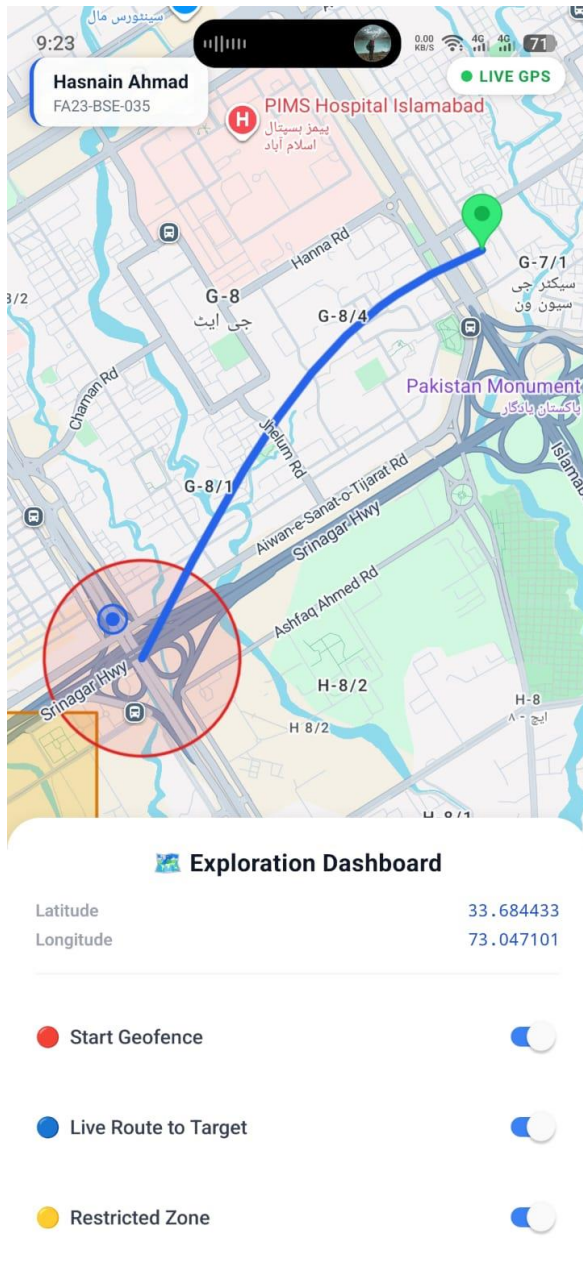
```
1 {
2   "expo": {
3     "name": "FA23-BSE-035",
4     "slug": "FA23-BSE-035",
5     "version": "1.0.0",
6     "orientation": "portrait",
7     "icon": "./assets/icon.png",
8     "userInterfaceStyle": "light",
9     "newArchEnabled": true,
10    "splash": {
11      "image": "./assets/splash-icon.png",
12      "resizeMode": "contain",
13      "backgroundColor": "#ffffff"
14    },
15    "ios": {
16      "supportsTablet": true
17    },
18    "android": {
19      "package": "com.fa23bse035.app",
20      "adaptiveIcon": {
21        "foregroundImage": "./assets/adaptive-icon.png",
22        "backgroundColor": "#ffffff"
23      },
24      "edgeToEdgeEnabled": true
25    },
26    "web": {
27      "favicon": "./assets/favicon.png"
28    }
29  }
30 }
```

Running App:

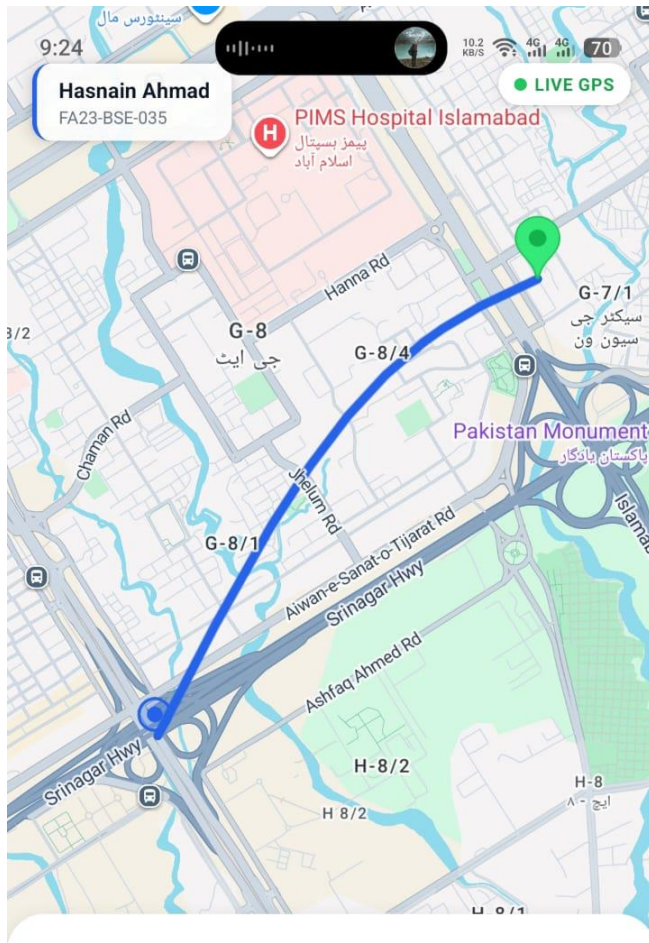
Exploring Different Components

Polygon:

Used to highlight specific geographic areas, territories.



Polyline: Use for route to specific area:



🗺️ Exploration Dashboard

Latitude 33.683694
Longitude 73.048275

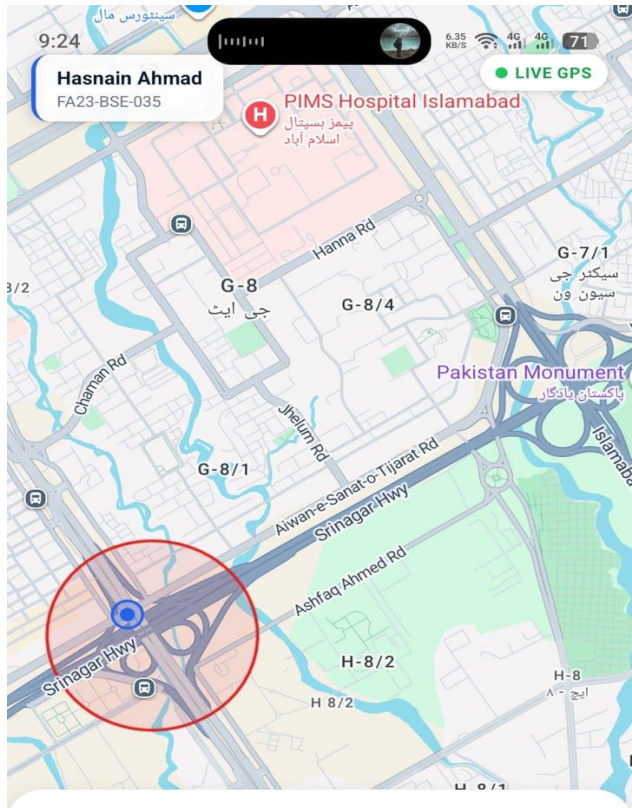
● Start Geofence ☐

● Live Route to Target ☒

● Restricted Zone ☐

Red Circle (Geofence):

- **The Purpose:** A **Geofence** defines a specific, measurable radius around a central point.



🗺️ Exploration Dashboard

Latitude 33.683794
Longitude 73.047378

🔴 Start Geofence ☒

🟦 Live Route to Target ☐

🟡 Restricted Zone ☐